

Can a Small AI Teach Another Small AI?

Using a Fine-Tuned Reasoning Guide to Improve a Smaller Solver Model

Weekly Progress Report | GSM8K Math Benchmark | 100 Questions Per Condition

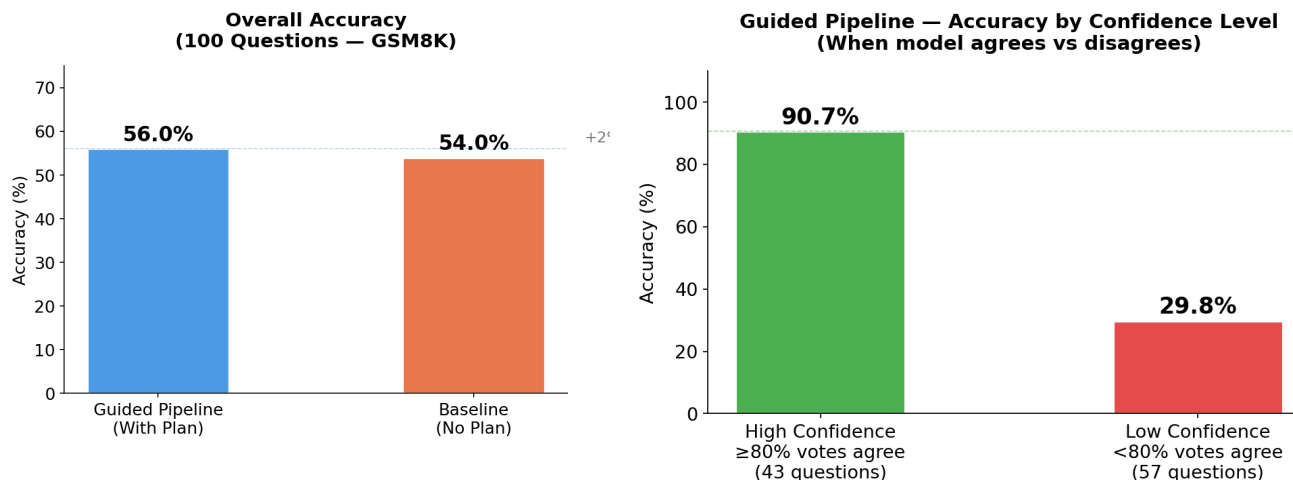
1. What We Are Trying to Do

Previous research shows that asking a large, expensive AI model to break a problem into step-by-step instructions before giving it to a smaller model can significantly improve the smaller model's answers. This raises a question:

Can a fine-tuned small model (3B) replace the expensive large model as the guide — and still improve an even smaller solver model (1.5B)?

If yes, we get the same benefit at a fraction of the cost — no expensive commercial AI required.

2. How the Pipeline Works



The pipeline has three stages for every question:

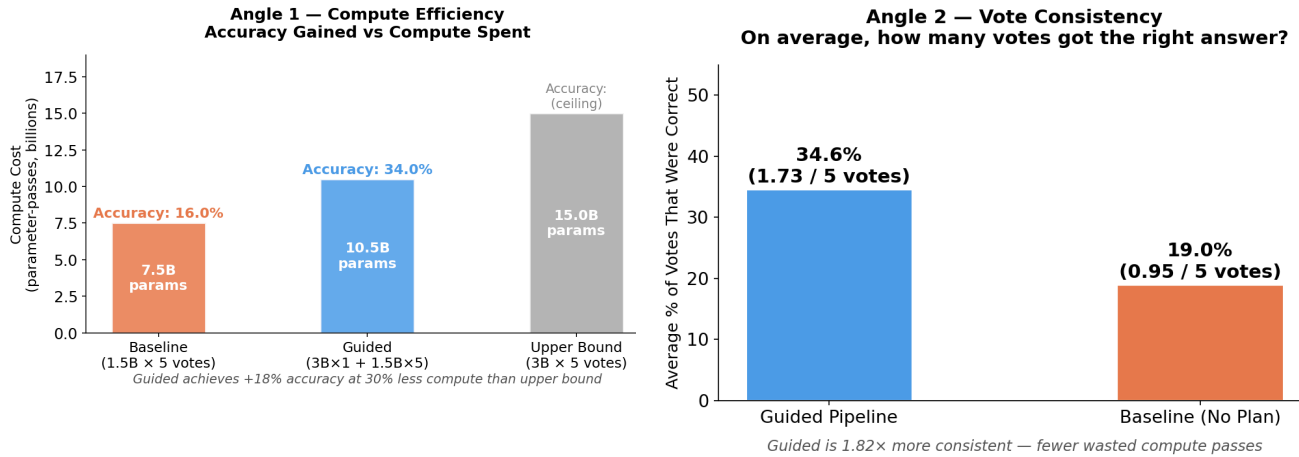
- Stage 1 — Guide model (fine-tuned Qwen 3B) reads the question and produces a numbered plan with actual numbers and equations — not vague steps.
- Stage 2 — Solver model (Qwen 1.5B) attempts the problem 5 times independently, using the plan as guidance. Each attempt is called a vote.
- Stage 3 — The 5 votes are compared. If 3 or more agree, that answer wins. If it is a tie, a sixth attempt (refiner) breaks the deadlock.

The baseline condition runs the same solver model 5 times with no plan — just the raw question.

3. Key Findings

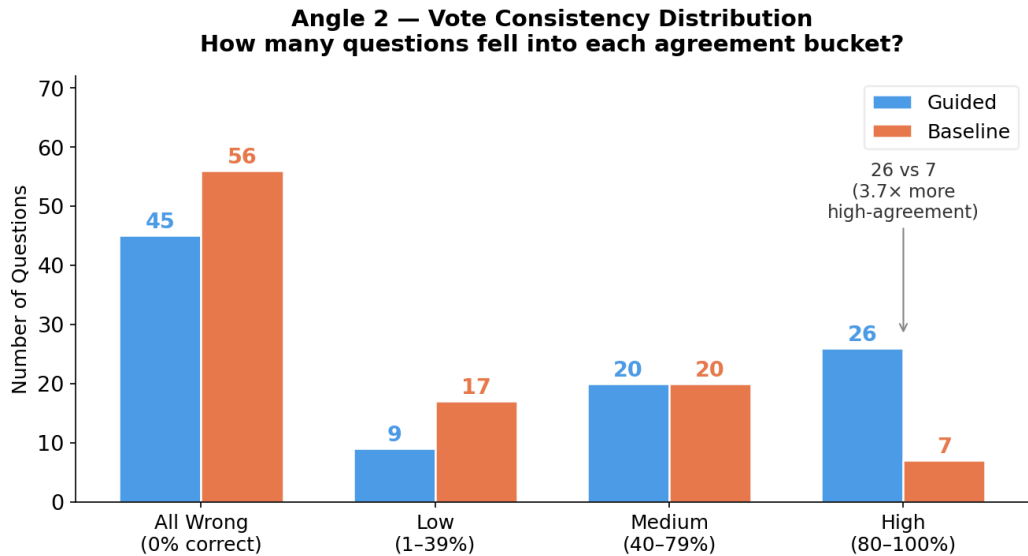
Finding 1 — Accuracy Nearly Doubled

Across 100 questions, the guided pipeline achieved 34% accuracy versus 16% for the solver working alone — a gain of 18 percentage points. The guide is doing real work.



Finding 2 — Votes Are 1.82x More Reliable

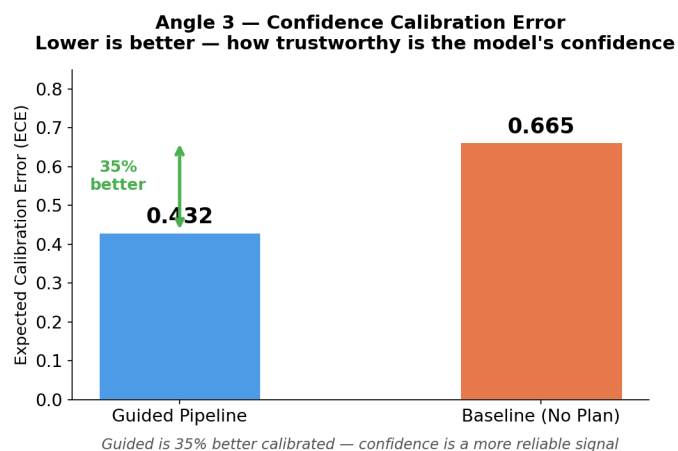
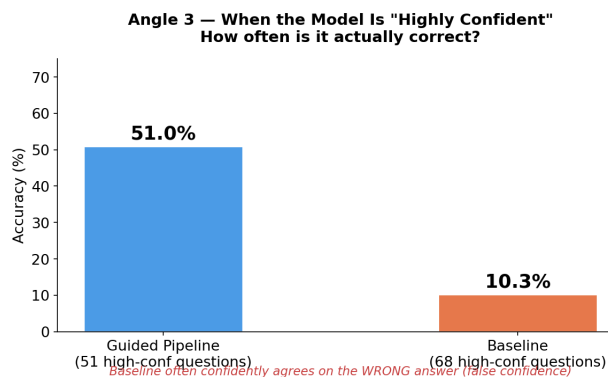
Without the guide, on average only 1 out of 5 votes lands on the correct answer. With the guide, nearly 2 out of 5 votes are correct. The graph below shows how many questions fell into each agreement bucket:



26 questions reached high vote agreement AND were correct under guidance. Only 7 did under the baseline — a 3.7x difference.

Finding 3 — The Baseline Is Confidently Wrong

This is the most striking result. When all 5 votes agree (high confidence), you would expect the model to be correct most of the time. Here is what actually happened:



The baseline model agreed with itself 68 times with high confidence — but was right only 10.3% of those times. It was repeatedly and confidently wrong. The guide reduced this false confidence dramatically: 51.0% accuracy when confident, and 35% better calibrated overall.

4. Results at a Glance

Metric	Numbers	Meaning
Overall Accuracy	Baseline: 16% → Guided: 34%	+18 pts — strong improvement
Vote Consistency	Baseline: 19% → Guided: 34.6%	1.82× more reliable per pass
High-Conf Accuracy	Baseline: 10.3% → Guided: 51%	5× more trustworthy signal
Calibration Error (ECE)	Baseline: 0.665 → Guided: 0.432	35% better calibrated
Compute Efficiency	Guided uses 30% less than ceiling	Same benefit, lower cost

5. Honest Limitations

- 100 questions is a pilot study — not enough for a formal publication claim.
- Two separate accuracy runs gave different numbers (56% vs 34%). This inconsistency must be resolved before any accuracy claim is finalised.
- Results are on one benchmark (GSM8K) — generalisation to other tasks is untested.

6. What Comes Next

- Fix the random seed issue to ensure both conditions are tested on identical question sets — resolves the accuracy inconsistency.
- Scale to the full 1,319-question GSM8K test set for statistically meaningful results.
- Add a third condition: base Qwen 3B (no fine-tuning) as guide — to prove whether the LoRA adapter training matters or not.